

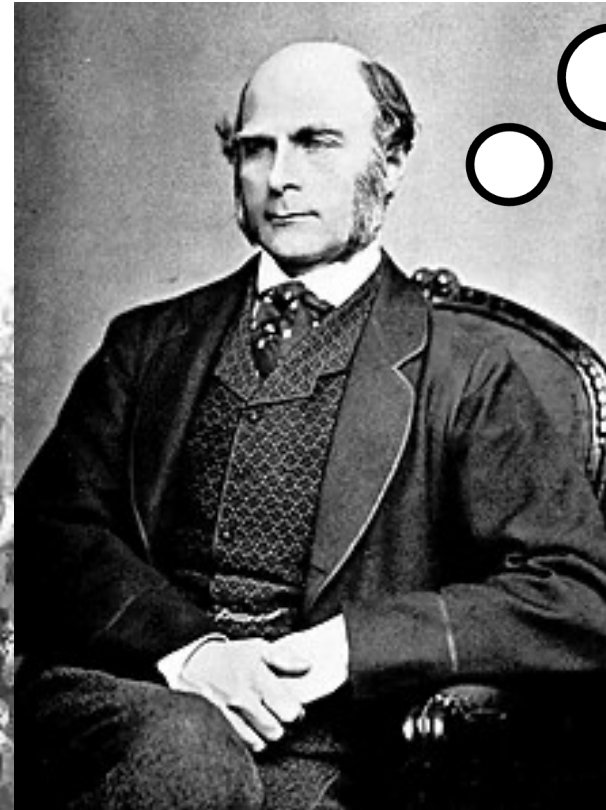
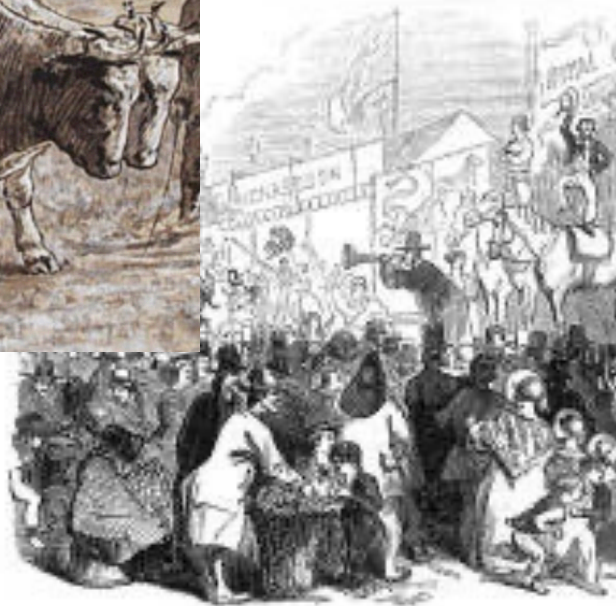
Self-censorship:

A measurement endeavour

November 13, 2024

Background:

GUESS THE WEIGHT



Well, we should use the
middlemost estimate
that expresses the vox
populi...

Wisdom of Crowds

SCIENCE ADVANCES | RESEARCH ARTICLE

SOCIAL SCIENCES

Scaling up fact-checking using the wisdom of crowds

Jennifer Allen^{1†}, Antonio A. Arechar^{1,2,3†}, Gordon Pennycook⁴, David G. Rand^{1,5,6*}

Professional fact-checking, a prominent approach to combating misinformation, does not scale easily. Moreover, some distrust fact-checkers because of alleged liberal bias. We explore a solution to these problems by using politically balanced groups of laypeople to identify misinformation at scale. Examining 207 news articles researched each article to those of 1128 Americans from Amazon Mechanical Turk who rated each line and lede. The average ratings of small, politically balanced crowds of laypeople (i) correlate with fact-checker ratings as well as the fact-checkers' ratings correlate with each other and (ii) predict the majority of fact-checkers rated a headline as "true" with high accuracy. Furthermore, cognitive reflection, knowledge, and Democratic Party preference are positively related to agreement with fact-checkers. Finally, rating each headline's publisher leads to a small increase in agreement with fact-checkers.

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Sage Journals

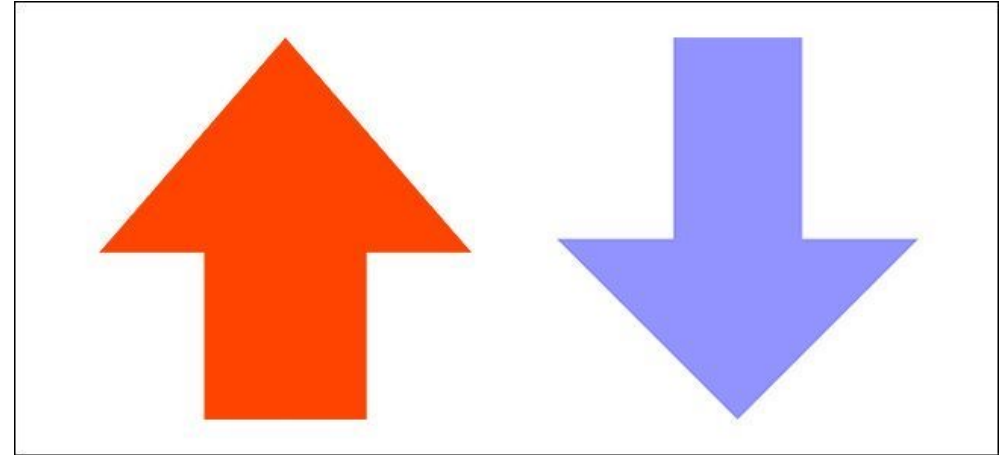
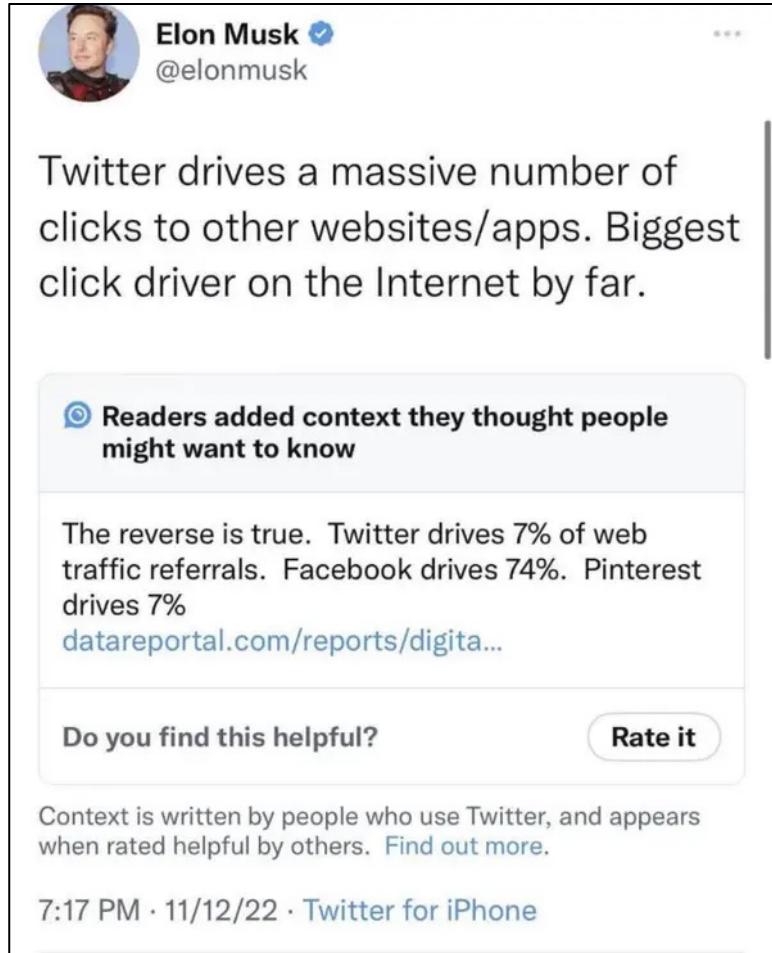
Crowds Can Effectively Identify Misinformation at Scale

Cameron Martel ¹, Jennifer Allen¹, Gordon Pennycook ², and David G. Rand ^{1,3,4}

Abstract

Identifying successful approaches for reducing the belief and spread of online misinformation is of great importance. Social media companies currently rely largely on professional fact-checking as their primary mechanism for identifying falsehoods. However, professional fact-checking has notable limitations regarding coverage and speed. In this article, we summarize research suggesting that the "wisdom of crowds" can be harnessed successfully to help identify misinformation at scale. Despite potential concerns about the abilities of laypeople to assess information quality, recent evidence demonstrates that aggregating judgments of groups of laypeople, or crowds, can effectively identify low-quality news sources and inaccurate news posts: Crowd ratings are strongly correlated with fact-checker ratings across a variety of studies using different designs, stimulus sets, and subject pools. We connect these experimental findings with recent attempts to deploy crowdsourced fact-checking in the field, and we close with recommendations and future directions for translating crowdsourced ratings into effective interventions.

Manifestations:



reddit

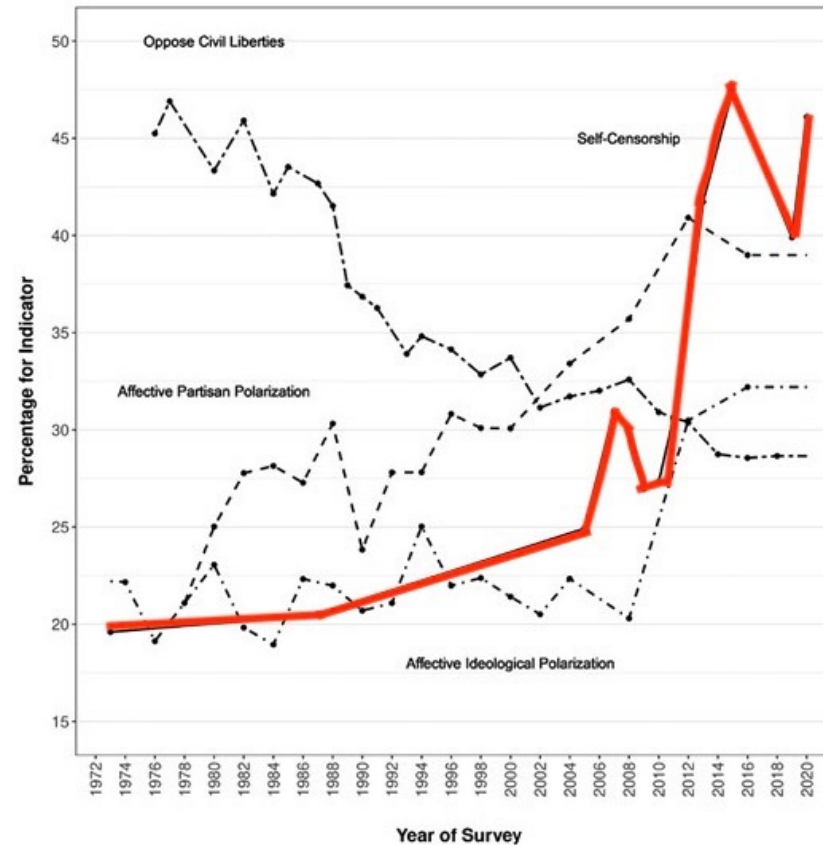


STACK OVERFLOW

Self-censorship:

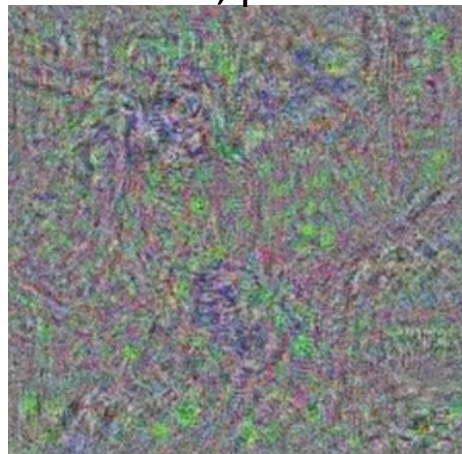
Withholding one's true opinion from an audience perceived to disagree with it

- Hayes et al., 2005



Who Self-censors?

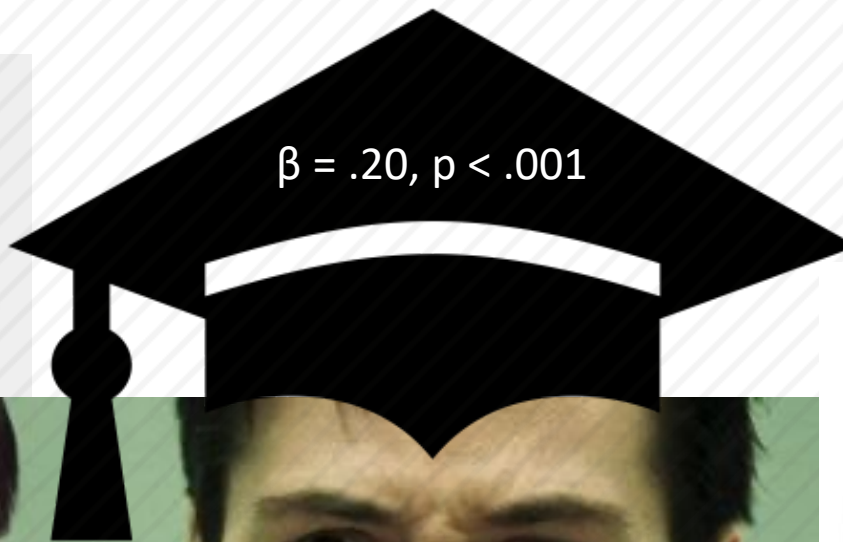
$r = .10, p = .025$



$\beta = .10, p = .018$



$\beta = .20, p < .001$



$\beta = .13, p = .019$

**FAKE
NEWS**



Measuring Self-censorship

“Thinking about the last few times you decided to not share your opinion (self-censor) on social media, what were the main reasons you considered when making those decisions?”

Scale Development

1. Themed OE responses
2. Created item pool (186 items -- themes, lit, and GPT)
3. Ran EFAs to establish scale
4. CFA validated the structure

4-factor Self-censorship Scale ($\alpha = .72$)

F1: Avoidance – 10 items ($\alpha = .87$)

“Silence is the best approach when discussing unpopular opinions”

F2: Confrontation/ Contrarian – 6 items ($\alpha = .79$)

“I secretly enjoy holding extreme positions”

F3: Knowledge – 4 items ($\alpha = .83$)

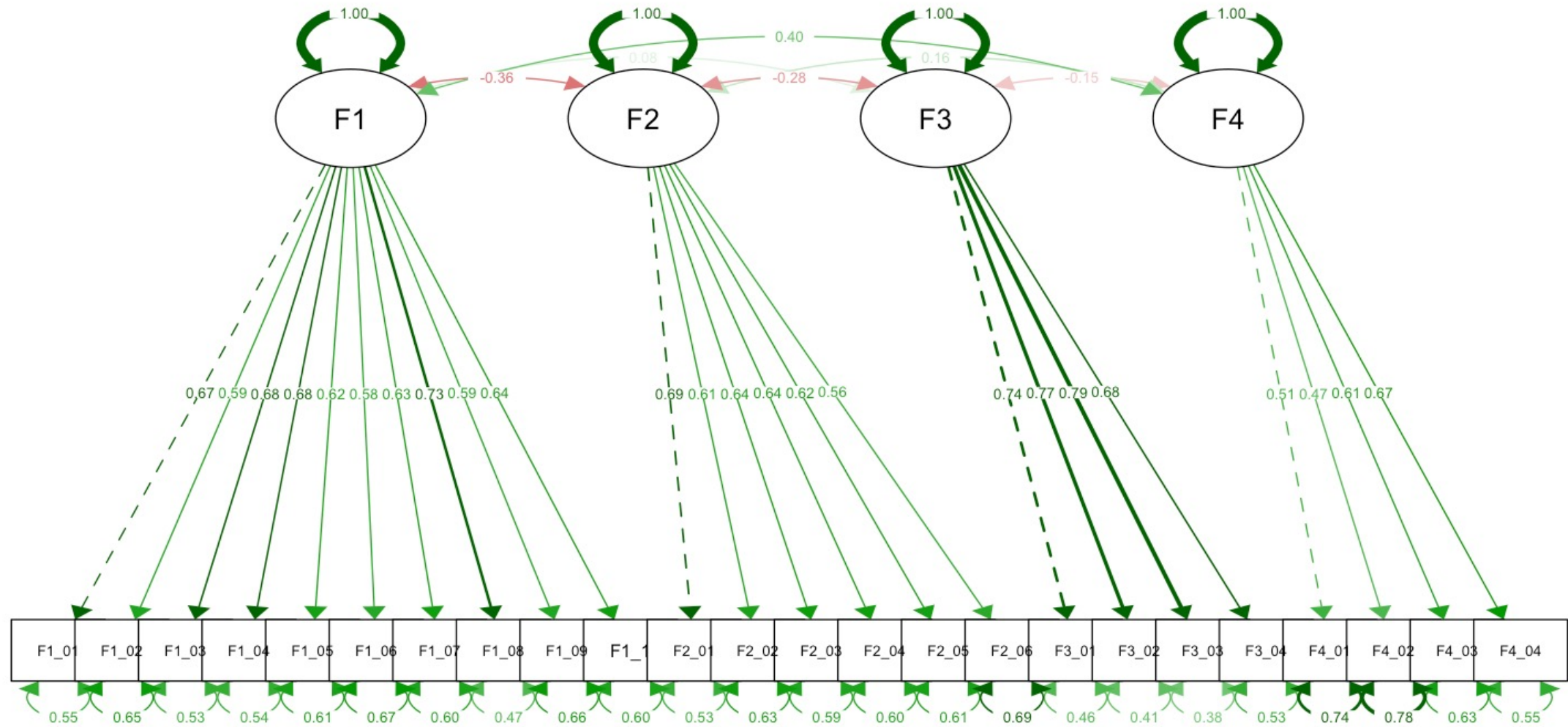
“I put a lot of care into the ideas I share”

F4: Conformity – 4 items ($\alpha = .66$)

“Mainstream opinions are the correct opinions”

4-factor self-censorship scale (alpha = .72) -- 24 items	
Factor 1 - Avoidance (alpha = .87) -- 10 items	
item10	I am more careful about what I say around people that hold different opinions
item27	Some topics are not safe to share or discuss
item30	I avoid topics that could hurt my reputation
item36	Silence is the best approach when discussing unpopular opinions
item54	It is not worth sharing my opinion if it could result in professional criticism
item69	Some topics are off-limits and I would never discuss them in public or online
item88	I will hide relevant information if it is uncomfortable to share
item132	I avoid contentious topics
item155	I often withhold my true opinions
item157	I subconsciously restrain from sharing extreme views
Factor 2 - Confrontation/ Contrarian (alpha = .79) -- 6 items	
item8	I share my opinion even when it could be offensive others
item20	It is ok to post morally questionable content online
item96	I am a confrontational person (e.g., I call it as I see it) regardless of the topic
item112	I secretly enjoy holding extreme positions
item143	Being careless with what I share or post online can be fun
item146	It's not my problem if my opinions hurt someone else
Factor 3 - Knowledge (alpha = .83) -- 4 items	
item64	Before speaking I consider how much I know about the topic
item78	I take steps to educate myself before discussing something online
item108	When I feel uncertain, I research the topic before posting
item150	I put a lot of care into the ideas I share
Factor 4 - Conformity (alpha = .66) -- 4 items	
item23	I would rather be politically correct than heard
item60	Before posting something online, I ask for a second opinion
item131	Mainstream opinions are the correct opinions
item137	I conform to others' opinions

F1 = Avoidance; F2 = Confrontation/ Contrarian; F3 = Knowledge; F4 = Conformity



Behavioural Measures

Psychological Review

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VOLUME 84 NUMBER 3 MAY 1977

Telling More Than We Can Know: Verbal Reports on Mental Processes

Richard E. Nisbett and Timothy DeCamp Wilson
University of Michigan

Evidence is reviewed which suggests that there may be little or no direct introspective access to higher order cognitive processes. Subjects are sometimes (a) unaware of the existence of a stimulus that importantly influenced a response, (b) unaware of the existence of the response, and (c) unaware that the stimulus has affected the response. It is proposed that when people attempt to report on their cognitive processes, that is, on the processes mediating the effects of a stimulus on a response, they do not do so on the basis of any true introspection. Instead, their reports are based on a priori, implicit causal theories, or judgments about the extent to which a particular stimulus is a plausible cause of a

Behavioural Measures Cont'd

- Willingness to share content
 - Anonymous vs not
- Petition
 - GPT generate a petition that they can sign AND share
 - Manipulate level of knowledge to tap into F3
- Correction paradigm
 - Confederate says something that contradicts researcher

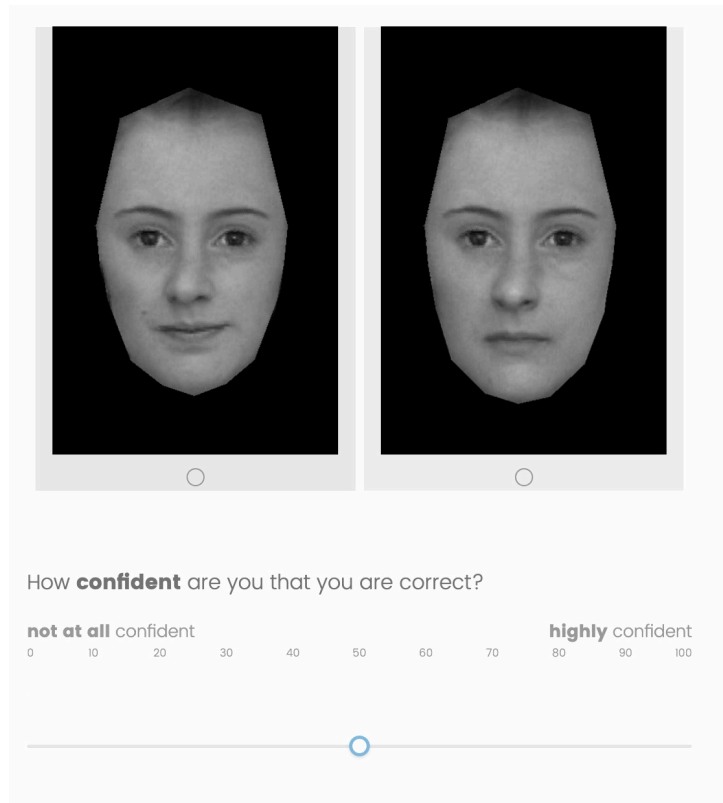
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Confidence Manipulations?

Study 1 – Cloud connect (N = 744)

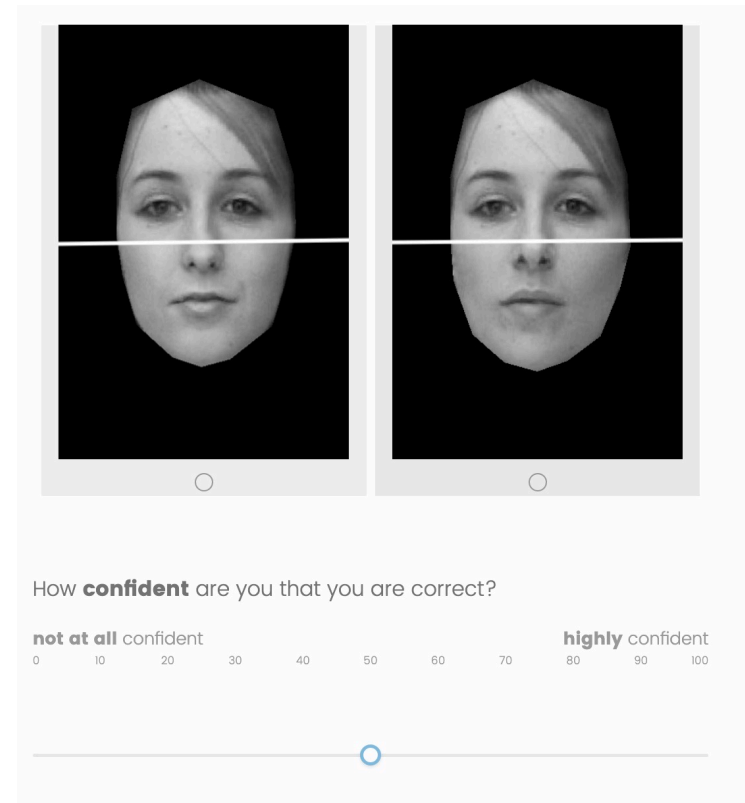
- Participants asked to identify the “real face” (10 trials/ block)
- Top and bottom morphs from different individuals (i.e., no correct answer)

Block 1



Block 1 trial interface. Two grayscale face images are shown side-by-side. Below each image is a small circle for selection. Below the images is a horizontal slider with a blue circle indicating a confidence level. The slider is labeled 'How confident are you that you are correct?' and has tick marks from 0 to 100. The left end is labeled 'not at all confident' and the right end is labeled 'highly confident'.

Block 2



Block 2 trial interface. Two grayscale face images are shown side-by-side. Below each image is a small circle for selection. Below the images is a horizontal slider with a blue circle indicating a confidence level. The slider is labeled 'How confident are you that you are correct?' and has tick marks from 0 to 100. The left end is labeled 'not at all confident' and the right end is labeled 'highly confident'.

Study 1 – Cloud connect (N = 744)

- Block 1: Received immediate feedback
 - Positive, negative, and neutral condition
- Block 2: No feedback
- Collected pre and post scores for confidence and placement

How **confident** are you that you will be able to correctly identify these faces?

not at all confident **highly** confident

0 10 20 30 40 50 60 70 80 90 100

_____○_____

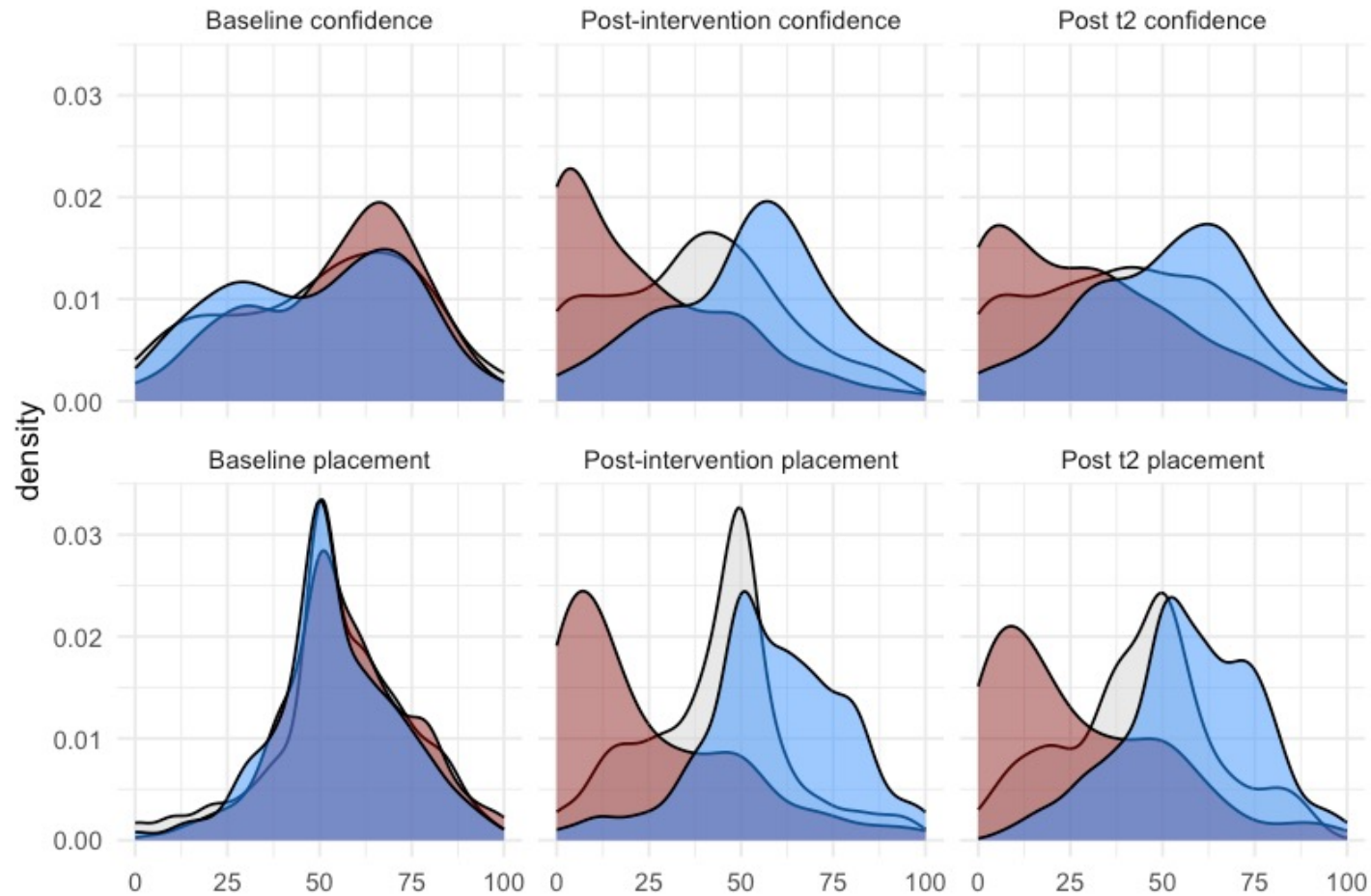
How well do you think you'll perform relative to **other participants** from Cloud connect?

worse than everyone **better** than everyone

0 10 20 30 40 50 60 70 80 90 100

_____○_____

Did the manipulation work?



Post ~ condition + baseline:
 $F(3, 709) = 204.9, p < .001$

Negative vs Control:
 $b = -16.03, p < .001$

Positive vs Control:
 $b = 15.90, p < .001$

Conceptual map for transfer effects?

- Types of tasks that would be interesting?
 - Perceptual
 - Knowledge base
 - Physical sensation
 - Personal judgments
 - Mystical/ uncanny experiences

Conceptual map for transfer effects?

- Similar tasks:
 - Blurred image task
 - Dot estimation task
 - Line length estimations
- Distant tasks:
 - General knowledge questions
 - Box jumping or light switch task
 - Colour matching task using RGB values